

Jafet Belmont, Sara Martino and Janine Illian

## 1 Welcome to the course!

- Welcome to the `inlabru` workshop!
- The aim of this workshop is to introduce you to a range of statistical modelling approaches, in particular the temporal, spatial and spatio-temporal modelling as implemented in the `inlabru` package.
- Workshop materials are available in the github repository

### Important!

Please use the instructions provided [here](#) to install and check your installation *before* the course start.

## 2 Learning Objectives for the workshop

At the end of the workshop, participants will have an understanding of:

- the motivation for and the challenges of analysing and modelling spatial data
- statistical models used to analyse spatial and spatio-temporal data
- the implementation of these models in the `inlabru` package
- how to independently analyse spatial data with `inlabru`

## 3 Intended audience

The workshop aims to cater for participants with a range of different backgrounds, who is interested in analysing data with modern spatial and spatio-temporal statistical modelling approaches.

## 4 Prerequisites

Participants should be familiar with the R environment, and general statistical approaches for modelling such as regression, analysis of (co)variance, and generalized linear models.

No knowledge of R-INLA or `inlabru` is required.

## 5 Schedule

**NOTE** The schedule is indicative and some changes might come!

### 5.0.1 Day 1

| Time          | Topic                                            |
|---------------|--------------------------------------------------|
| 10:00 - 10:30 | Welcome                                          |
| 10:30 - 11:30 | <b>Session 1:</b> Introduction to inlabru        |
| 11:30 - 13:00 | <b>Session 2:</b> Some practical tips on inlabru |
| 13:00 - 14:30 | Lunch break ☐                                    |
| 14:30 - 15:30 | <b>Practical Session 1</b>                       |
| 15:30 - 15:45 | Coffee Break ☐                                   |
| 15:45 - 16:45 | <b>Practical Session 2</b>                       |
| 16:45 - 17:00 | wrap-up and outlook                              |

### 5.0.2 Day 2

| Time          | Topic                                                |
|---------------|------------------------------------------------------|
| 9:00 - 10:00  | <b>Session 3:</b> Time Series Models                 |
| 10:00 - 10:30 | Snack ☐                                              |
| 10:30 - 11:30 | <b>Session 3 cont.:</b> Time Series Models           |
| 11:30 - 13:00 | <b>Practical Session 3</b>                           |
| 13:00 - 14:30 | Lunch break ☐                                        |
| 14:30 - 15:30 | <b>Session 4:</b> Introduction to Spatial Statistics |
| 15:35 - 15:45 | Coffee Break ☐                                       |
| 15:45 - 16:45 | <b>Practical Session 4</b>                           |
| 16:45 - 17:00 | wrap-up and outlook                                  |

### 5.0.3 Day 3

| Time          | Topic                                  |
|---------------|----------------------------------------|
| 9:00 - 10:00  | <b>Session 5:</b> Areal processes      |
| 10:00 - 10:30 | Snack ☐                                |
| 10:30 - 11:30 | <b>Session 6:</b> Geostatistics        |
| 11:30 - 13:00 | <b>Practical Session 5</b>             |
| 13:00 - 14:30 | Lunch break ☐                          |
| 14:30 - 15:30 | <b>Session 7:</b> Point Processes      |
| 15:35 - 15:45 | Coffee Break ☐                         |
| 15:45 - 16:45 | <b>Practical Session 5 (continued)</b> |
| 16:45 - 17:00 | wrap-up and outlook                    |

### 5.0.4 Day 4

| Time          | Topic                                             |
|---------------|---------------------------------------------------|
| 9:00 - 10:00  | <b>Session 8:</b> Spatiotemporal models           |
| 10:00 - 10:30 | Snack ☐                                           |
| 10:30 - 11:30 | <b>Session 9:</b> Model comparison and evaluation |
| 11:30 - 13:00 | <b>Practical Session 6</b>                        |
| 13:00 - 14:30 | Lunch break ☐                                     |

| Time          | Topic                                                       |
|---------------|-------------------------------------------------------------|
| 14:30 - 15:30 | <b>Session 10: Multi-likelihood/joint likelihood models</b> |
| 15:35 - 15:45 | Coffee Break ☐                                              |
| 15:45 - 16:45 | Practical Session 7                                         |
| 16:45 - 17:00 | Closing session                                             |